Assignment

Module 2.3 Peripheral Troubleshooting Theory

1. **List 5 common issues users face with USB keyboards or mice on laptops and write one possible solution for each.**

- **Prevalent USB Keyboard and Mouse Complications and Remedies:**

1) USB Device Unrecognized: The laptop does not identify the keyboard or mouse, resulting in complete inability to utilize the peripheral device.
   Resolution: Detach the device carefully and reinsert it into an alternative USB port on the computer. If the issue persists, attempt using different USB ports sequentially to identify a functional connection point.
2) Keyboard Keys Unresponsive: Certain keyboard keys do not function as intended, preventing proper text input or command execution.
   Resolution: Reattach the keyboard firmly to its USB port and verify that the keys operate correctly by testing multiple keys across different sections of the keyboard. Consider cleaning the keyboard's connections if problems continue.
3) Mouse Malfunction: The mouse cursor remains stationary on the display, making navigation and selection tasks impossible.
   Resolution: Attempt connecting the mouse to a different USB port on the computer. Verify that the mouse sensor is clean and unobstructed, as dust or debris may impede proper functionality.
4) Recurrent Disconnection: The keyboard or mouse experiences repeated disconnections during operation, causing intermittent loss of functionality and workflow interruption.
   Resolution: Inspect the USB cable thoroughly for visible deterioration, fraying, or damage. Ensure the connection is firmly established at both the device and computer ends and consider replacing the cable if damage is evident.
5) Sluggish or Deferred Performance: The keyboard or mouse exhibits delayed responsiveness to user input, creating a frustrating lag between action and system response.
   Resolution: Power cycles the laptop completely by shutting down and restarting the system. Additionally, upgrade the device drivers as required through the manufacturer's website or system settings to ensure optimal compatibility and performance.

2. **Simulate a troubleshooting flow: Your Bluetooth headphones are not connected to your phone. Write the exact step-by-step actions you would take to diagnose and fix the issue, mentioning any settings or indicators you would check.**

- If my Bluetooth headphones fail to establish a connection with my phone, I would implement the following systematic approach to diagnose and resolve the issue.

Step 1: Verify the headphone battery level

I would ensure that the headphones are adequately charged by examining the battery indicator or charging light to confirm sufficient power availability.

Step 2: Activate Bluetooth functionality on the phone

I would navigate to Settings → Bluetooth and confirm that Bluetooth is enabled.

Step 3: Initiate pairing mode on the headphones

I would depress and maintain pressure on the headphones' power or Bluetooth button until the indicator light begins to flash, typically displaying blue or red coloration depending on the device model.

Step 4: Review the list of available devices

I would examine the available devices list within the phone's Bluetooth settings to determine whether the headphone name is displayed.

Step 5: Select the headphones from the device list

I would select the headphone name and await confirmation that the phone displays a Connected or Paired status.

Step 6: Disconnect competing devices

Should the headphones be connected to another phone or laptop, I would terminate those connections and attempt the pairing process anew.

Step 7: Remove and re-establish the pairing

If the headphones remain unconnected, I will locate the headphones in the Bluetooth settings, select Forget or Unpair, and subsequently repeat the pairing procedure.

Step 8: Perform a restart of both devices

I would power off the headphones and restart the phone. Subsequently, I would reactivate the headphones and attempt reconnection.

Step 9: Validate the connection quality

Following a successful connection, I would play audio content and confirm that sound is transmitted through the headphones. I would additionally verify the phone's audio output configuration and volume level.

Step 10: Execute a factory reset of the headphones if required

Should the issue persist, I would adhere to the manufacturer's specifications to reset the headphones and undertake the pairing process once more.

3. **A user reports that their external webcam is not detected during a Zoom call on Windows 10. Write a checklist of 6 things to check or try to resolve the problem.

Hint: Consider device manager, privacy settings, and app permissions. **

- **Check your USB connection:** Make sure the webcam is plugged firmly to your computer. If your computer doesn't recognize it, plug it into a different USB port.
  Look in Device Manager: Open Device Manager and find Cameras or Imaging devices. If your webcam is turned off or has a warning sign, turn it on or install the latest driver.
  Check your camera privacy settings: Go to Settings → Privacy → Camera and turn on camera access.
  Allow apps to use your camera: In Windows camera privacy settings, turn on Allow desktop apps to access your camera so Zoom can use the webcam.
  Pick the right camera in Zoom: Open Zoom, go to Settings → Video, and pick your external webcam from the list.
  Restart and test: Close any other programs using the webcam, restart your computer, and test if the camera works in Zoom.

4. **You are helping a friend whose printer is showing 'offline' status on their PC. Describe how you would use the Windows Troubleshooter tool to try to fix the issue and explain what information the tool provides during the process.**

- When a printer displays an “Offline” message on a Windows PC, I would use the Windows Troubleshooter to identify the cause and attempt to fix it.
  I would open Settings → Devices → Printers & scanners, select the affected printer, and check for a troubleshooting option. Another way is to go to Settings → Update & Security → Troubleshoot → Additional troubleshooters → Printer and run the troubleshooter.
  Windows would examine the printer for common issues, such as faulty connections, incorrect settings, outdated drivers, or problems with the print spooler service. It would show the errors found, display diagnostic information, and suggest ways to solve them. If possible, Windows may also repair the detected problems automatically.
  After completing the recommended fixes, I would restart the printer if needed and print out a test page to verify that it is working properly.